Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 104606

Subject: Digital CMOS VLSI Design

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only) **[2 x 7 = 14]**

(a) Which is the first step in CMOS fabrication?
(i) Lithography (iii) Diffusion
(ii) Oxidation (iv) Ion implantation

(b) nWell process is commonly used in:
(i) PMOS fabrication (iii) CMOS fabrication
(ii) NMOS fabrication (iv) Bipolar fabrication

(c) In a MOS transistor, the current between drain and source flows due to the movement of:
(i) Electrons only (iii) Electrons and holes
(ii) Holes only (iv) Neither electrons nor holes

(d) What controls the conductivity in a MOSFET?
(i) Source voltage (iii) Gate voltage
(ii) Drain voltage (iv) Substrate

(e) Which of the following loads is used in a CMOS inverter?
(i) Resistor (iii) Enhancement-mode MOSFET
(ii) Depletion-mode MOSFET (iv) Capacitor

(f) Inverter delay time can be minimized by:
(i) Increasing supply voltage (iii) Both a and b
(ii) Reducing load capacitance (iv) None of the above

(g) Which of the following gates is a basic building block of CMOS logic?
(i) AND gate (iii) XOR gate
(ii) NOT gate (iv) NAND gate

(h) The logic level of a MOS circuit is determined by:
(i) Gate voltage (iii) Drain voltage
(ii) Source voltage (iv) Substrate voltage

(i) In MOS logic circuits, a pass gate is constructed using:
(i) Only PMOS (iii) Both NMOS and PMOS
(ii) Only NMOS (iv) None of the above

(j) What does MOS stand for in MOSFET?
(i) Metal Oxide Semiconductor (iii) Metal Organic Semiconductor
(ii) Metal Oxide Silicon (iv) Metal Oxide Structure

Q.2 (a) Explain the structure and operation of a MOSFET with a neat diagram. **[7]**
(b) How does the nWell process differ from the pWell process in CMOS technology? **[7]**

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Q.3	(a)	Describe the current-voltage characteristics of MOSFETs in different operating regions.	[7]
	(b)	Describe the CMOS fabrication process in detail, highlighting each step.	[7]
Q.4	(a)	Compare the operation of n-channel and p-channel MOSFETs.	[7]
	(b)	Explain the importance of mask layers in CMOS fabrication and their design considerations.	[7]
Q.5	(a)	What is switching power dissipation in CMOS inverters? Derive the formula and explain the factors affecting it.	[7]
	(b)	Explain the working principle of a CMOS NAND gate with a circuit diagram.	[7]
Q.6	(a)	Explain the voltage transfer characteristics of a CMOS inverter.	[7]
	(b)	Design a 4-input CMOS logic circuit for the AND-OR-INVERT function and explain its working.	[7]
Q.7	(a)	Discuss the factors affect the delay time of a MOS inverter and how it can be optimized.	[7]
	(b)	Explain the role of clock signals in sequential logic circuits and its impact on circuit performance.	[7]
Q.8	(a)	Discuss the advantages and disadvantages of MOS logic circuits having depletion-mode nMOS loads.	[7]
	(b)	Design a clocked CMOS D flip-flop and explain its operation.	[7]
Q.9	(a)	Explain the operation of a CMOS SR latch with a detailed circuit diagram.	[7]
	(b)	Describe how interconnect parasitics impact the performance of MOS inverters.	[7]

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